

Vibe Analysis

E-commerce Business Performance Analysis Report

Project Scope: Raw Data → Data Profiling → Data Cleaning → Star Schema → Business Analysis → Interactive HTML Dashboard

Executive Summary

The Vibe Analysis project demonstrates an end-to-end Data Analyst workflow for an e-commerce business. Four raw datasets were profiled, cleaned, transformed into an analytical model, and used to generate business insights and an interactive HTML dashboard. The final analytical dataset contains **25,000 unique orders**, **2,000 customers**, **100 products**, and **731 unique dates** covering January 2024 to December 2025.

The business generated approximately **■29.50 million in revenue** and **■6.76 million in profit**, with an overall profit margin of **22.9%**. North was the highest-profit region, Home & Kitchen was the highest-revenue category, and several high-revenue products were identified as having weak profit margins.

1. Project Objective

The objective was to transform raw e-commerce data into a structured analytical solution capable of answering business questions related to revenue, profit, orders, customers, products, categories, regions, and monthly performance.

2. Source Data

Dataset	Description	Records / Scope
Orders	Order-level transactional data including sales, cost, profit, quantity, customer, product and region references.	25,050 raw rows
Customers	Customer attributes including demographics, segment and signup date.	2,010 raw rows
Products	Product, category, sub-category, cost and selling price information.	100 products
Regions	Geographic mapping of region, state and city.	20 records

3. Data Quality Assessment

Orders Data

Issue	Finding
Exact duplicate transactions	50

Issue	Finding
Invalid order dates	10
Missing customer IDs	60
Missing product IDs	35
Negative quantity transactions	15
Profit calculation mismatches	15

Customers Data

The customer dataset contained 10 duplicate records. These were removed to create a dimension with one row per customer, resulting in 2,000 unique customers.

Products Data

Category values contained inconsistent capitalization and trailing spaces. Examples included 'electronics', 'OFFICE SUPPLIES', and a Furniture value with trailing whitespace. Categories were standardized to Clothing, Electronics, Furniture, Home & Kitchen, and Office Supplies.

4. Data Cleaning and Transformation

Customers: Duplicate customer IDs were removed, resulting in a clean Dim_Customers table with 2,000 unique customers.

Products: Category formatting was standardized and potentially inconsistent category/sub-category combinations were flagged for business review.

Orders: 50 exact duplicate transactions were removed. Invalid dates were retained as missing values, missing customer and product IDs were flagged, and negative quantities were flagged using an is_return indicator.

Date Dimension: A Dim_Date table with one row per valid date was created, containing date_key, date, day, month, month_name, quarter, and year.

5. Analytical Data Model

A star schema was proposed with Fact_Orders as the central transactional fact table. The dimension tables are Dim_Customers, Dim_Products, Dim_Regions, and Dim_Date.

Table	Role	Key Business Content
Fact_Orders	Fact	Sales amount, cost, profit, quantity, discount and transaction keys
Dim_Customers	Dimension	Customer name, gender, age, segment and signup date
Dim_Products	Dimension	Product, category, sub-category, cost and selling price
Dim_Regions	Dimension	Region, state and city

Table	Role	Key Business Content
Dim_Date	Dimension	Date, day, month, month name, quarter and year

6. Overall Business Performance

KPI	Result
Total Revenue	■29.50 Million
Total Profit	■6.76 Million
Overall Profit Margin	22.9%
Unique Orders	25,000
Unique Customers	2,000
Return Transactions	15

7. Regional Performance

Region	Revenue	Profit	Profit Margin
North	■7.57M	■1.72M	22.7%
South	■7.38M	■1.70M	23.0%
East	■7.35M	■1.70M	23.1%
West	■5.78M	■1.33M	23.0%
Central	■1.42M	■0.32M	22.3%

Key insight: North generated the highest total profit. Central significantly underperformed in total revenue and profit, although its profit margin was broadly similar to other regions. This suggests lower sales volume rather than fundamentally weaker unit profitability.

8. Category Performance

Category	Revenue	Profit	Profit Margin
Home & Kitchen	■6.76M	■1.60M	23.6%
Clothing	■6.54M	■1.53M	23.4%
Office Supplies	■6.04M	■1.28M	21.1%
Furniture	■5.83M	■1.49M	25.6%

Category	Revenue	Profit	Profit Margin
Electronics	■4.29M	■0.86M	20.0%

Key insights: Home & Kitchen generated the highest revenue. Furniture achieved the highest category profit margin, while Electronics had the lowest margin among the major categories and should be reviewed for pricing, discounting, and cost issues.

9. Top Product Performance

Rank	Product	Revenue	Profit Margin
1	Kids Product 98	■829K	38.5%
2	Tables Product 97	■728K	29.0%
3	Storage Product 83	■699K	26.4%
4	Appliances Product 99	■695K	32.9%
5	Appliances Product 50	■668K	30.3%
6	Cookware Product 59	■658K	29.9%
7	Kids Product 80	■641K	25.2%
8	Women Product 67	■635K	22.8%
9	Chairs Product 12	■569K	34.6%
10	Paper Product 94	■523K	15.1%

Kids Product 98 was the strongest product by revenue and also maintained a strong profit margin, making it a highly valuable product from both a sales and profitability perspective.

10. High Revenue but Low Profit Margin Products

Product	Revenue	Profit	Profit Margin
Cookware Product 89	■501K	■48K	9.6%
Mobiles Product 66	■417K	■50K	12.1%

These products generate strong revenue but weak profitability. They should be investigated for high costs, excessive discounts, low pricing, or supplier-related cost increases.

11. Monthly Performance

The analysis covers 24 months from January 2024 through December 2025. Revenue remained broadly stable throughout the period.

Metric	Best Period	Result
Highest Monthly Revenue	December 2025	■1.33M
Highest Monthly Profit	August 2025	■303K

An important observation is that the month with the highest revenue was not the month with the highest profit. This confirms that revenue alone should not be used as the primary measure of business performance.

12. Growth Assessment

Year-over-year performance was relatively stable. Revenue increased by approximately 0.4%, while profit increased by approximately 0.2%. The business is financially stable but demonstrates limited growth momentum.

13. Dashboard Solution

An interactive HTML dashboard was developed to support business users and portfolio presentation.

KPIs: Total Revenue, Total Profit, Total Orders, Unique Customers

Visuals: Monthly Revenue and Profit Trend, Revenue by Category, Profit by Region, and Top 10 Products by Revenue

Filters: Year, Region, and Category

All dashboard KPIs and charts update dynamically based on the selected filters, enabling interactive exploration of business performance.

14. Key Business Recommendations

- 1. Improve low-margin products.** Review Cookware Product 89 and Mobiles Product 66 for pricing, cost, and discount optimization.
- 2. Investigate Electronics profitability.** Conduct product-level margin analysis to identify products driving the category's lower profitability.
- 3. Improve Central region performance.** Investigate customer demand, marketing effectiveness, market coverage, and product availability.
- 4. Invest in high-margin opportunities.** Furniture offers the strongest category margin, while Home & Kitchen offers the highest revenue.
- 5. Maintain data quality controls.** Future data pipelines should automatically validate duplicates, invalid dates, missing foreign keys, category inconsistencies, and returns.

15. Conclusion

Vibe Analysis successfully demonstrates a complete Data Analyst portfolio workflow from raw business data to decision-ready insights. The project showcases data profiling, data cleaning, dimensional modeling, KPI analysis, profitability analysis, trend analysis, and interactive dashboard development.

The business generated strong overall revenue and maintained a healthy 22.9% profit margin. The primary opportunities for improvement are expanding the Central region, improving Electronics profitability, optimizing high-revenue low-margin products, and prioritizing high-margin product categories.

Final Outcome: A complete analytical solution consisting of cleaned dimension and fact tables, a date dimension, business insights, and an interactive HTML dashboard.